Open questions from Grothendieck's Montpellier notes: fourteen statements a formalisation could not reach

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Working paper, 25 October 2026

Abstract

The University of Montpellier has put online 178 folders of Alexandre Grothendieck's mathematical working notes. Since August 2026 an edition has transcribed 164 of them with a language model, restated each in current notation, searched the literature for the statements they contain, and formalised what it could in Lean 4 against Mathlib. Most of the statements that look new sit in theories Mathlib does not have yet: étale cohomology, topoi, stacks, abelian schemes, motives. This note collects fourteen of them for mathematicians. Each comes with its folder and page, the context, what the literature search found, the argument the page gives or sketches, the steps the edition added, and what would settle it. None of the statements has been checked by a specialist, and several may already be in print; a reference is as welcome as a proof or a counterexample.

1 Why this note

Grothendieck gave his mathematical papers to Jean Malgoire in 1990 and 1995. They were deposited at the University of Montpellier in 2010 and given to it in 2012, and the university placed 178 folders, 16,074 pages by its inventory's count, in open access as facsimiles. They are working notes: drafts in ink and pencil, restarted, struck, annotated, almost never dated. Twenty folders have been edited by hand. The rest had not been transcribed.

Since August 2026 an edition has read the remaining folders with a language model as first transcriber, and written for each folder a *modernised reading*, a restatement in current notation made from the transcription and never from the hand. A register of *findings* lists the statements of the folders that may not stand in the literature, each written as a claim about the literature with what was searched and what would settle it; at the time of writing it holds 463 rows, of which 81 turned out to be in print. Statements were then formalised in Lean 4 against a pinned revision of Mathlib, about two thousand theorems in all. A companion paper describes the formalisation [17].

Lean settles a statement only when the objects it speaks of exist in the library. In October 2026, 203 open findings with no proof were sorted by what Mathlib can reach: 16 were reachable as it stands, 39 in part, and 147 not at all, because they live in étale or crystalline cohomology, topoi, stacks, derivators, motives or Hodge theory [17]. Those are exactly the parts of the notes where Grothendieck's own work was. The aim of this note is to hand the

most promising of them to people who can prove them with pen and paper, or say where they are already proved.

How to read the entries

Each problem gives its *source*: the folder (a link to Montpellier's facsimile), the pages in the archive's pagination, and the inventory's dating, whose brackets mark an inferred date. « The page » means what Grothendieck wrote, as transcribed; short quotations are his. « The reading » means the edition's modernised reading, made by a language model. « We » marks a step of the edition: an argument, a correction or a search made for this note or for the register, with a language model, and not checked by a specialist unless the entry says so. The entries make no claim of priority. A statement on an undated leaf says when he wrote it down and nothing about whether others had.

The transcriptions themselves are not public: the rights belong to Grothendieck's heirs, and the reading views stay off the site until they authorise them. The facsimiles are public, at Montpellier, and each entry cites the page to look at. Montpellier's servers were down at the end of October 2026; the links follow the archive's own scheme and should work again.

2 Étale cohomology and Hodge theory

Problem 1 (A global bound with a shift of 3)**.** For a field k of characteristic ℓ put $\ell^{\nu(k)} = [k : k^{\ell}]$, and for a scheme Z of characteristic ℓ let $\nu(Z)$ be the supremum of $\nu(k(z))$ over its maximal points. Let X be an affine scheme on which ℓ is not invertible, so that $V(\ell) \subset X$ has characteristic ℓ. Is it true that

$$\operatorname{cd}_\ell(X) \;\le\; \sup\Big(\sup_{x\in X} \operatorname{cd}_\ell k(x),\; \nu(V(\ell)) + 3\Big)\,?$$

Source. folder 31, p. 84, *SGA A [SGA 4] (résidus de rédaction)*, inventory dating [1963–1973]. The page is headed « Conjectures sur la cohomologie des schémas affines »; this is its conjecture 2°).

Context. M. Artin wrote in SGA 4 X 3.0 that in unequal characteristic one must « certainement tenir compte » of $[k : k^{\ell}]$, without a bound. The same page states a local form, 4°): for U open in the spectrum of a henselian local domain A of residue characteristic ℓ with fraction field of characteristic 0, $\operatorname{cd}_\ell(U) \le \dim A + \nu(k) + \operatorname{cd}_\ell(k)$. The words around « ouvert » in 4°) are hard to read, so whether U is meant to be affine, and whether it may meet $V(\ell)$, is not known.

What is known. The local form holds, in a sharper form, for U a non-empty affine open of $\operatorname{Spec} A[1/\ell]$ with A excellent: Gabber and Orgogozo prove $\operatorname{cd}_\ell(U) = \dim A + \dim_\ell(k)$ [14, Th. 6.1], and Kato's ℓ-dimension satisfies $\dim_\ell(k) \le \nu(k) + \operatorname{cd}_\ell(k)$. Mathew bounds cd_ℓ of affine opens of $\operatorname{Spec} R[1/\ell]$ for R excellent normal and henselian along an ideal containing ℓ, in terms of $\dim R_{\mathfrak{p}} + \log_\ell[\kappa(\mathfrak{p}) : \kappa(\mathfrak{p})^{\ell}]$ [23, Prop. 4.5]. Globally we found only $\operatorname{cd}_\ell(X) \le \dim X + \sup_x \operatorname{cd}_\ell k(x)$ for X qcqs of finite dimension [6, Cor. 3.29], which gives $2n+3$ for $\operatorname{Spec} \mathbb{Z}_\ell[x_1, \dots, x_n]$ where the page's bound gives $n+3$. Gabber and Zavyalov settle the analogous expectation for affinoids [15, Th. 1.7]; the Travaux de Gabber treat ℓ invertible

only.

Ideas. The constant cannot be lowered to 2 (our observation): for $X = \operatorname{Spec}\mathbb{Z}$ and ℓ odd, $\nu(V(\ell)) = 0$ and $\sup_x \operatorname{cd}_\ell k(x) = \operatorname{cd}_\ell \mathbb{Q} = 2$, while Artin–Verdier duality gives $H^3(\operatorname{Spec}\mathbb{Z}, j_!\mu_\ell) \cong \mathbb{Z}/\ell$ [24, II.3], so $\operatorname{cd}_\ell(\operatorname{Spec}\mathbb{Z}) = 3$. A natural route is the excision sequence for $V(\ell) \subset X$ and its complement $X[1/\ell]$: $V(\ell)$ is affine of characteristic ℓ, so of ℓ-cohomological dimension at most 1 [28, X 5.1]; local cohomology along $V(\ell)$ can be moved to the henselisation along $V(\ell)$ by Gabber's affine analogue of proper base change, where Gabber–Orgogozo and Mathew apply. The difficulty is to absorb the dimension terms of those local bounds into $\sup_x \operatorname{cd}_\ell k(x)$, which the page's bound allows, and to see where the shift 3 comes from: the page does not explain it.

Lean. Not attempted: Mathlib has no étale cohomology.

What would settle it. A printed bound of this shape for an arbitrary affine scheme of mixed characteristic, or a counterexample to the 3.

Problem 2 (Lower bound for a local domain: the exceptions)**.** Let A be a noetherian local domain with fraction field K and residue field k, and r a prime. The page states that

$$\operatorname{cd}_r(K) \;\geq\; \operatorname{cd}_r(k) + \dim A$$

with exactly two possible exceptions: (a) r is the characteristic of K; (b) $r = 2$ and some point x of $\operatorname{Spec} A$ has an orderable residue field $k(x)$ with $\operatorname{cd}_2 k(x)(\sqrt{-1})$ finite. It adds that exception (b) disappears when A is regular, and reduces to « k itself real » when A is catenary and universally japanese. Are these refinements true?

Source. folder 31, p. 22, inventory dating [1963–1973]. The margin gives the example that keeps exception (b) alive: the local ring at the origin of $\mathbb{R}[X,Y]/(X^2+Y^2)$, a real domain whose only real point is the origin. The margin also asks: « cette restriction est-elle essentielle ??? »

What is known. The inequality is SGA 4 X 2.4 when r is invertible in A, with the cruder exception « some residue field orderable », and with the page's marginal example [28]. When r is the residue characteristic and $\operatorname{char} K = 0$, it follows for A excellent henselian from Gabber–Orgogozo §3 [14]. We found nothing on the finer exception (b), on the regular case, on the catenary refinement, or on r the residue characteristic for A neither excellent nor henselian. Serre's *Cohomologie galoisienne* II §4 was not consulted.

Ideas. Artin's proof goes by induction on $\dim A$, down to a henselian discrete valuation ring through a height-one prime [28, X 2.3]. The finer exception (b) says that what matters is the virtual cohomological dimension of the real residue fields, which suggests running the same induction with cd_2 replaced by cd_2 of $k(x)(\sqrt{-1})$ at the real points. The page's exception list covers r equal to the residue characteristic in mixed characteristic, which SGA 4 X 2.4 does not.

Lean. Not attempted (Galois cohomology of fields is in Mathlib only in low degree).

What would settle it. A reference for (b) and the two refinements, or an example of a regular local domain where the inequality fails at $r = 2$.

Problem 3 (Hodge types of the deleted neighbourhood of a point)**.** Let Y be a complex algebraic variety of dimension n, $j : X \hookrightarrow Y$ an open immersion with X smooth and dense, $y \in Y \smallsetminus X$, and $H^i = H^i(X \cap B_y) = (R^i j_* \mathbb{Q})_y$ with its mixed Hodge structure [9]. Then

$$h^{p,q}(H^i) \neq 0 \implies p, q \in [\max(0, i-n+1),\ \min(i, n)],$$

so the weights lie in $[0, 2i]$ for $i \leq n-1$ and in $[2(i-n+1), 2n]$ for $i \geq n$, and the level $\max|p-q|$ is at most $\min(i, 2n-1-i) \leq \min(i, n-1)$. Is this in the literature, and what is the right relative statement for a proper morphism?

Source. folder 12, pp. 115–119, *Généralités (sous-groupes de Galois motiviques)*, inventory dating 1964–[c. 1971]. Page 119 boxes « poids $\leq 2\inf(i, n)$, niveau $\leq \inf(i, n-1)$ » for X regular, $F = \mathbb{Q}_\ell$ and f an immersion with $n = \dim \overline{f(X)}$, in his yoga of weights and levels of ℓ-adic sheaves. The same page asks, with a question mark, for a relative version with $n = d_y(f)$ and level $\leq \min(i, n)$. The typescript says of its own conclusions (p. 96): « la portée des « résultats » de notre travail est nécessairement sujette à caution, puisqu'ils dépendent de conjectures non prouvées à l'heure actuelle » (the Weil conjectures and resolution of singularities).

Context. The page's argument is a Leray spectral sequence for a factorisation of f as an open immersion with normal crossings complement followed by a proper map, conditional on resolution, purity and the Weil conjectures, with tables of weights and levels. The Hodge-theoretic statement above is our translation.

What is known. Deligne bounds the Hodge types of the cohomology of a variety [8, 8.2.4]; Durfee constructs the mixed Hodge structure on $H^i(X \cap B_y)$ and its weight spectral sequence; Durfee–Saito, Steenbrink, Navarro Aznar and El Zein bound weights; for isolated singularities the box is implicit in the literature on links, for instance [13, §2]. We did not find the box, or the level bound, for arbitrary y.

Ideas. Our argument: take a resolution $\pi : \tilde{Y} \to Y$, an isomorphism over X, with $D = \pi^{-1}(Y \smallsetminus X)$ simple normal crossings; put $F = \pi^{-1}(y)$ and $F_q = F \cap D^{(q)}$. Then $\dim F_0 \leq n-1$ and $\dim F_q \leq n-q$. Durfee's weight spectral sequence $E_1 = H^{k-q}(F_q)(-q) \Rightarrow H^k(X \cap B_y)$ and Hodge III 8.2.4 (i), (ii) for the compact F_q put every E_1 term in the box. Locally the lower end moves up by one because $\pi^{-1}(y)$ has dimension at most $n-1$. The cone over a smooth projective V of dimension $n-1$ with $h^{i,0}(V) \neq 0$ attains level i for $i \leq n-1$ and $n-1$ at $i = n$. For a proper f with fibre dimension d the gain fails, since a smooth family of curves already has H^1 of level $1 = d$; so the relative question of page 119 cannot keep $n-1$ if $d_y(f)$ is the fibre dimension.

Lean. Not attempted: no Hodge structures on cohomology in Mathlib.

What would settle it. A reference for the box; a relative statement interpolating between the immersion case and the smooth proper case; its ℓ-adic form with level read as divisibility of Frobenius eigenvalues.

Problem 4 (An Ext^2 that vanishes when 2 is invertible?)**.** Is $\mathcal{E}xt^2_{\mathbb{Z}}(G, M) = 0$, on the fppf site of a base on which 2 is invertible, for every finite locally free commutative group scheme G and every quasi-coherent M?

Source. folder 129, p. 95, *Notes séminaire Montréal*, inventory dating [1970]. A parenthetical NB after a theorem on Ext over truncated rings: « $\mathrm{Ext}^2_{\mathbb{Z}}(G,M)=0$ si 2 inv., pour tt G fini loc. libre », with the remark that this simplifies the deformation theory of Barsotti–Tate groups for $p\neq 2$. The word « inv. » is uncertain in the transcription, and the page gives no proof.

What is known. Illusie places the obstruction to deforming a truncated Barsotti–Tate group in $\mathrm{Ext}^2_{\mathbb{Z}}(G_0, t_{G_0}\otimes J)$ and says that proving directly that it vanishes is out of reach [18]. On $S_0=\operatorname{Spec} k$ the NB, read this way, would kill that obstruction at once for p odd. That is a tension with Illusie's account, not a refutation.

Ideas. The first test is $G=\alpha_p$, $M=\mathbb{G}_a$ over a perfect field of odd characteristic, with Breen's computations of the extensions of the additive group [3]. If $\mathcal{E}xt^2(\alpha_p,\mathbb{G}_a)\neq 0$, the NB as read is false and the facsimile should be checked for another word than « inv. »

Lean. Not attempted: no fppf cohomology in Mathlib.

What would settle it. The computation of $\mathcal{E}xt^2(\alpha_p,\mathbb{G}_a)$ for p odd, or a reference for the vanishing.

3 Algebraic cycles

Problem 5 (Splitting the critical Lefschetz condition)**.** Let X be smooth projective of dimension n over an algebraically closed field, with a Weil cohomology H, a smooth hyperplane section $\varphi: Y\hookrightarrow X$, and L the Lefschetz operator. Write H^{2i} for the cohomology in which codimension-i cycles live, and consider:

- $A'(X)$: for $2i\leq n-1$, a class $x\in H^{2i}(X)$ whose restriction $\varphi^* x$ is algebraic is algebraic;
- $A''(X)$: for $2i\geq n-1$, every algebraic class of $H^{2i+2}(X)$ is $\varphi_* y$ for an algebraic $y\in H^{2i}(Y)$;

and their critical versions A'^0, A''^0, with $2i=n-2$ for n even and $2i=n-1$ for n odd. Let $A^0(X)$ be the critical Lefschetz condition: $L^2: H^{n-2}(X)\to H^{n+2}(X)$ for n even, $L: H^{n-1}(X)\to H^{n+1}(X)$ for n odd, is a bijection on algebraic classes. Under which hypotheses do $A'^0(X)$ and $A''^0(X)$ imply $A^0(X)$, and is the resulting reduction of the Lefschetz standard conjecture,

$$C(X)\iff C(Y)+A'^0(X\times X)+A''^0(X\times X),$$

correct as stated?

Source. folder 16, p. 49, *Formalisme algébrique des correspondances et algèbres de Poincaré-Hodge*, inventory dating 1966–1967, in a letter of January 1967. The letters use A, C, D in senses that do not match the later published labels; the conditions above are the reading's restatement. The letter to Coates places the question in the motivic picture (p. 48): « The complete yoga then being that we have the category of motives which is semi-simple ! ».

What is known. The equivalence $C(X)\iff C(Y)+A^0(X\times X)$ and its descent along a chain of hyperplane sections are Kleiman's [20, Th. 2.13]. The splitting of A^0 into the two one-sided conditions was not found in Kleiman 1968 and 1994, Grothendieck 1969, Charles 2013, Ramón Marí, or Milne; André's book [1, ch. 5] remains to be read.

Ideas. The page factors the critical operator through the section. For $n = 2m$: $L^2 = \varphi_* \circ L_Y \circ \varphi^* : H^{2m-2}(X) \to H^{2m-2}(Y) \to H^{2m}(Y) \to H^{2m+2}(X)$; for $n = 2m+1$: $L = \varphi_*\varphi^* : H^{2m}(X) \to H^{2m+2}(X)$. With it (our reconstruction, $n = 2m$): an algebraic $z \in H^{2m+2}(X)$ is $\varphi_* y$ with y algebraic by A'^0; by weak and hard Lefschetz, $y = L_Y \varphi^* x$ for a unique x, and $z = L^2 x$; if L_Y^{-1} preserves algebraic classes on Y, $\varphi^* x$ is algebraic, and then x is algebraic by A'^0. So the splitting holds given the Lefschetz condition on the section in the relevant degree. In the displayed equivalence it is applied to $T = X \times X$, whose hyperplane section is not $Y \times Y$; whether the hypothesis on the section is supplied by the induction, or must be added, is the point to check.

Lean. Not attempted: no Weil cohomology in Mathlib.

What would settle it. A reference for the splitting, or a precise statement of the hypotheses under which it holds, applied to $X \times X$.

4 Abelian varieties and reductive groups

Problem 6 (Polarised abelian surfaces with even Euler characteristic)**.** Let A/S be an abelian scheme of relative dimension 2 with dual B, and δ a section of $\mathrm{NS}_{A/S}$ with $\chi(\delta)$ even. Is the torsor $\mathrm{Pic}^{\delta}_{A/S}$ under B trivial? Over a field k, this says that the Poonen–Stoll class $c_\delta \in H^1(k, B)$ vanishes [25].

Source. folder 59, pp. 11–13, *Construction des faisceaux inversibles sur schémas de Picard*, inventory dating [années 1960–1970]. Page 11 gives a canonical isomorphism $\mathrm{Pic}^{\delta'}_{B/S} \simeq \tilde{\delta}'_* \mathrm{Pic}^{\delta}_{A/S}$; page 13 composes it with its analogue and concludes, for $n = 2$ and $\chi(\delta)$ even, that Pic^{δ} is trivial. That last line rests on an uncertain « si » and an illegible word.

What is known. Poonen and Stoll define c_λ and prove $2c_\lambda = 0$, with no criterion in terms of $\chi(\lambda)$ or the dimension [25, §4]. We found no such criterion elsewhere.

Ideas. The reading thought the conclusion depends on compatibilities of trivialisations that the folder asks for and never proves. Our argument, unchecked, needs none for mere triviality. The map $L \mapsto \det(\hat{L})^{\pm 1}$, with $\hat{L}$ the Fourier–Mukai transform, is a morphism of fppf sheaves $\mathrm{Pic}^{\delta}_{A/S} \to \mathrm{Pic}^{\varphi(\delta)}_{B/S}$, equivariant along $\varphi(\delta)^\sim$ by the theorem of the square. The same construction on B goes back to $\mathrm{Pic}^{\varphi\varphi(\delta)}_{A/S}$, and in dimension 2, $\varphi\varphi(\delta) = \delta$. The composite $B \to B$ is multiplication by $\mp\chi(\delta)$. Hence $(\chi(\delta) \pm 1)\,[\mathrm{Pic}^{\delta}] = 0$, and $2\,[\mathrm{Pic}^{\delta}] = 0$ through $L \mapsto L \otimes [-1]^* L$. Both $\chi \pm 1$ are odd when χ is even, so the class vanishes whatever the signs. Whether the trivialisation obtained is canonical is a separate question.

Lean. Not attempted: no abelian schemes in Mathlib.

What would settle it. A specialist's check of the argument. One abelian surface over a number field with a polarisation of type $(1, 2)$ and $c_\lambda \neq 0$ refutes the statement. A sharp consequence to test: for a genus-2 curve C/k whose Jacobian has real multiplication by $\mathcal{O}_D$, $D \equiv 1 \bmod 8$, defined over k, the theta class is a sum of two classes of even χ, so the statement would force Pic^1_C to have a k-point.

Problem 7 (Counting Kostant rigidifications)**.** Let G be a simple adjoint group with Coxeter number h, Weyl group W, and z the order of the centre of its simply connected cover, over a field of characteristic p with $(h,p)=1$. Consider quadruples (T,B,T',B') with T, T' maximal tori in apposition in Kostant's sense and $B \supset T$, $B' \supset T'$ Borel subgroups. Does G act freely, with quotient finite étale of rank $|W|^2/(h\,z\,\varphi(h))$, and of rank $|W|^2/(h\,z)$ once a generator of $T \cap N(T') \simeq \mu_h$ is fixed?

Source. folder 123, pp. 21, 28, 30, 34, *[Autour de Kostant]*, letter to Kostant of 22 October 1969 (p. 21) and p. 34. The letter's footnote describes $N(T)\cap N(T')$ as an extension of $(\mathbb{Z}/h)^*$ by $\mathbb{Z}/h \times \mathfrak{z}$, of order $h\,z\,\varphi(h)$. No proof is given.

Ideas. The number is an integer in the small cases: $1, 2, 4, 12$ for A_1, A_2, B_2, G_2. Our step: an element fixing (T,B,T',B') lies in $N(T)\cap B = T$ and in T', so the action is free once $T \cap T' = 1$, which page 30 writes. The count then reduces to the order of $N(T)\cap N(T')$, whose image in W should be the extension of $P^\vee/Q^\vee$ by $(\mathbb{Z}/h)^*$.

What is known. We found no such count. Kostant 1959 and Springer's *Regular elements of finite reflection groups* (1974) are where it would be.

Lean. Not attempted in Lean. A computer count over $\mathbf{F}_q$, for primes $q \equiv 1 \bmod h$ up to 37 (`computations/kostant`), finds a free action with 2, 4 and 12 orbits for A_2, B_2 and G_2, and 4, 8 and 24 once a generator is fixed, agreeing with $|W|^2/(hz\varphi(h))$, and finds $|N(T)\cap N(T')| = hz\varphi(h)$ in each case.

What would settle it. A proof of $|N(T)\cap N(T')| = h\,z\,\varphi(h)$, or the count in print.

5 Topoi and stacks

Problem 8 (Two classifying topoi that are not coherent)**.** The theories of commutative rings with irreducible spectrum (nilradical prime) and of rings with zero-dimensional spectrum are geometric. The page says their classifying topoi, subtopoi of the coherent classifying topos of rings, are not coherent, their presenting topologies on affine schemes of finite type over $\mathbb{Z}$ not being quasi-compact. Is that right, and do these topoi have enough points?

Source. folder 161-2, pp. 39, 41–42, *Algèbre universelle*, inventory dating [c. 1963–1973]. Page 39: « Cette topologie n'est pas quasi-compacte »; page 42, margin: « topologie non cohérente ! A-t-elle assez de pts ?? »; page 41, margin: « encore un topos non cohérent (bien que sous-topos d'un topos cohérent) ».

What is known. Coherent toposes have enough points (Deligne); these theories are geometric but, as presented, use the infinite disjunction hidden in « nilpotent ». We found neither non-coherence nor the existence of points stated for them, in Johnstone or in Caramello's work on theories of rings [19, 5].

Ideas. Non-coherence needs more than the presentation: one must show that no coherent axiomatisation exists, for instance by exhibiting a cover in the classifying topos with no finite

subcover. For points, Barr's theorem gives a Boolean cover, not points; a model-theoretic argument, or an explicit set of local rings detecting the theory, is needed.

Lean. Not attempted: classifying topoi are not in Mathlib.

What would settle it. A proof or reference for non-coherence, and an answer to the question about points.

Problem 9 (Foliations and their quotient topoi)**.** Is there an equivalence between the category of pairs (X, R), with X a topological space and R a local equivalence relation on X, locally open with locally connected fibres, and the category of triples $(X, \mathcal{Y}, f)$, with $f : \mathrm{Top}(X) \to \mathcal{Y}$ a locally open geometric morphism with locally connected fibres to a « multiplicité topologique », a 2-category whose Hom-categories are discrete?

Source. folder 119, pp. 144–145, dated « Mars 1986 » on p. 144. Under « Th »: « On a $C \rightleftarrows C'$ », both arrows marked $\approx$. No proof. A third condition on f is partly illegible; the reading suggests it may be a simple connectedness of the fibres.

What is known. SGA 4 IV, Exercise 9.8.3 j)–k), gives the functor $(X, R) \mapsto (\mathrm{Top}(X) \to \mathrm{Top}(X/R))$ [28]; Rosenthal proves $\mathrm{Top}(X/R)$ a topos for strictly open R [27]; Kock and Moerdijk construct the inverse, $a \mapsto \mathrm{Lker}(a)$, and show the round trips are identities under simple connectedness or kernel-pair conditions [21, Lemma 5.1, §6], and characterise étendues as $\mathrm{sh}(M, r)$ [22]. We found no statement of an equivalence of categories.

Ideas. The two functors are there; the problem is the right conditions on both sides so that they are mutually inverse, and which quotient topos the page means: the topos of R-invariant sheaves (sheaves on a monodromy groupoid) or the holonomy quotient, which Kock and Moerdijk distinguish.

Lean. Not attempted: no geometric morphisms of topoi in Mathlib.

What would settle it. A correct statement with proof, or Moerdijk's Zaragoza lectures (1993), which we could not read and where it might be.

Problem 10 (Realising the symmetry invariant of a Picard stack)**.** Let M, N be abelian sheaves on a topos, and ${}_2N$ the 2-torsion of N. Is every homomorphism $M \to {}_2N$ the symmetry invariant of a Picard stack pinned by M and N?

Source. folder 134-1, p. 5, letter to Deligne, Lodève, 27 August 1974 (Breen's copy, forwarded by Deligne): « Je sais prouver (sauf erreur) que tout hom. $M \to {}_2N$ provient d'un champ de Picard convenable (épinglé par M, N) (a priori l'obstruction est dans $\mathrm{Ext}^3(X; M, N)$, mais un argument « universel » prouve qu'elle est nulle) ». No argument is written. On p. 6 the letter asks his own version of this problem: « Question : Le triangle exact (T) et la suite exacte (*) sont-ils connus par les compétences (Quillen, Breen, Illusie…) ? Connaissent-ils des variantes « supérieures » ? » (are they known to the experts, and do they know higher variants?).

What is known. Picard stacks with invariants M, N are classified by $\mathrm{RHom}(\mathbb{Z}(M), N)$ truncated in degree 2 (*Pursuing Stacks*; Breen 1999 [4]). No printed source states the realisation, or refutes it on a topos.

Ideas. Two sketches disagree. The reading realises the invariant by hand for M free, on the split stack $M \times BN$ with an antisymmetric form b such that $b(e_i, e_i) = q(e_i)$, and passes to general M through a resolution. We doubt that step, from memory and without having checked it: the construction depends on a basis. On the classifying topos of abelian groups, a Picard stack pinned by the generic group U and $U/2U$, with the projection as invariant, would make Moore spectra truncated above π_1 functorial in finitely presented abelian groups; but the stable Moore-space category on finitely generated groups is a non-split linear extension, split on cyclic groups [2, Rem. 1.5.5]. If that translation is right, the obstruction does not vanish universally. The statement does hold over a point and for M constant and free.

Lean. Not attempted: no stacks on a topos in Mathlib.

What would settle it. The differential $\mathrm{Hom}(M/2, N) \to \mathrm{Ext}^3(M, N)$ of the universal coefficient spectral sequence on the classifying topos of abelian groups, or Grothendieck's « universal » argument written out.

Problem 11 (Morphisms of ramification domains are fully faithful)**.** Over a scheme S, a « domaine de ramification » is a stack of multigaloisian categories over the étale site of S with a continuous geometric realisation into relative schemes, subject to axioms R1 and R2 (pp. 190–193). For a domain (R, r) defined by a scheme Z with a finite group G of operators, is $\mathrm{Hom}((R, r), (R', r'))$ equivalent to the category of pairs (P', α), P' a galoisian object of R'_S of group G and α a G-isomorphism $r'(P') \simeq Z$, so that every homomorphism of domains is fully faithful?

Source. folder 29, p. 197, *Groupe fondamental [autour de SGA 4 et SGA 7]*, inventory dating 1959–1969: Proposition 6, stated without proof, with the corollary « nécessairement pleinement fidèle ». The axiom R2 of pp. 190–193 is itself not verified by the edition. Earlier in the folder (p. 24) he notes where removing the auxiliary choices leads: « il semble qu'on soit infailliblement conduit à prendre comme catégorie [...] la catégorie [...] elle-même qu'on se propose de construire et d'étudier ! », comparing it with sites and their topoi.

What is known. Nothing found. The work on stacks of ramified Galois covers (*Stacks of ramified Galois covers*, arXiv:1307.1116; *Ramified Galois covers via monoidal functors*, arXiv:1507.05309) is where an equivalent statement would sit; neither surfaced in our search, which is not the same as their not containing it.

Ideas. The corollary says that a ramification domain maps into another only as a subdomain. The proof should reduce to the case (R, r) given by (Z, G), where a morphism is determined by the image of the universal G-torsor.

Lean. Not attempted: no étale site or stacks of categories in Mathlib.

What would settle it. A proof of Proposition 6 from R1–R2, or a reference.

6 Moduli of curves and Galois actions

Problem 12 (The Galois image, seen on one curve)**.** Let X be an anabelian curve over a field K of finite type over $\mathbb{Q}$, and $\hat{\Pi}_1(X, I)$ its profinite fundamental groupoid on a finite set I of base points. The page lists necessary conditions for a vertex-fixing automorphism Θ of $\hat{\Pi}_1(X, I)$ to come from $\mathrm{Gal}(\bar{K}/K)$: it lifts to $\hat{\Pi}_1(X', I')$ for every finite étale cover X' of X defined over K; it preserves the kernels of $\hat{\pi}_1(X', x') \to \hat{\pi}_1(Y, f(x'))$ for every K-morphism $f : X' \to Y$; it is compatible with morphisms between covers; and it satisfies the loop condition at the cusps with a single multiplier. Are these conditions sufficient?

Source. folder 142, pp. 94–95, 98–101, *Action de Galois sur Teichmüller*, inventory dating [from 1978]. Page 95 conjectures the conditions sufficient « lorsque X est (disons) une courbe algébrique anabélienne, avec K extension de type fini de $\mathbb{Q}$ »; page 98 calls the loop condition « sûrement la plus importante de toutes, et de très loin ! ». Pages 98–99 are largely illegible.

What is known. Ihara's question and the Oda–Matsumoto conjecture characterise the Galois group as the automorphisms of the geometric fundamental group functor on all varieties compatible with all morphisms; Pop proved it in that form, and tripod variants for finite étale covers of $\mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$ [26]. Hoshi–Mochizuki and others give injectivity for a single hyperbolic curve; Collas's survey leaves a combinatorial candidate for $G_{\mathbb{Q}}$ open [7]. We found no characterisation attached to one curve and its covers, with base points and the inertia condition. Pop's tripod paper could not be read.

Ideas. If Pop's tripod variant covers the covers of one hyperbolic curve, the problem reduces to comparing his compatibility conditions with the page's list; the loop condition would then be the extra information that makes one curve suffice.

Lean. Not attempted: no étale fundamental group in Mathlib.

What would settle it. A proof of sufficiency, a counterexample, or a published variant that implies it.

Problem 13 (The deleted neighbourhood of a normal crossings divisor)**.** Let X be a smooth complex analytic space (or orbifold) with a normal crossings divisor Θ, and $V^*_{\Theta,X}$ the deleted tubular neighbourhood of Θ. Is $V^*_{\Theta,X}$ the homotopy colimit of the semi-simplicial system of deleted tubular neighbourhoods of the strata of the unfolding of Θ, so that its fundamental groupoid is the amalgamated sum $\tilde{\Pi}_1 \leftarrow \tilde{\Pi}_{2,1} \to \tilde{\Pi}_2$ of the groupoids of the torus bundles over the open strata of codimension 1 and 2? And hence the same for $\Pi_1(X \smallsetminus \Theta)$ whenever $X \smallsetminus \Theta$ and its end have equivalent fundamental groupoids?

Source. folder 147, pp. 148–151, *Structure à l'infini des* $M_{g,\nu}$. The folder is undated; the inventory dates its group (141–149) [from 1978]–1983. We date it to the winter of 1982–1983, because its formalism is the one Grothendieck describes to Illusie in a letter of 22–25 January 1983 (IHÉS, fonds Grothendieck–Illusie, 1ILL35, p. 27) and in a *rapport d'activité* of which he sent Illusie a copy on 29 March 1983 (1ILL41, p. 4; copy in folder 119, pp. 141–142). Page 149 states a « Th. de recollement » as an equivalence of topoi and corrects it at once: « en

fait, c'est une équivalence d'homotopie ». Pages 150–151 give the corollaries, with a purity argument for neglecting the strata of codimension 3. No proof is written.

Context. The folder describes the boundary of $\overline{M}_{g,\nu}$ by stable graphs, lists the strata of modular dimension at most 2, and writes on its p. 29 (archive p. 59) that a more or less complete understanding of the discrete π_1 in these small cases « donnera la clef pour une compréhension complète dans le cas général »: the two-level principle of the Teichmüller tower, about a year before the *Esquisse d'un programme*. Applied to $X \smallsetminus \Theta = M_{g,\nu}$, the corollaries would express the mapping class group through the strata of codimension 1 and 2; that application is the reading's, not the page's.

What is known. The two-level principle for the discrete tower was proved by Hatcher, Lochak and Schneps [16]. For the general gluing statement, the places to look are the real oriented blow-up and the Kato–Nakayama space of the log structure, whose boundary is a torus-bundle model of the deleted neighbourhood, and Looijenga's and Boggi's work on the boundary of $\overline{M}_{g,n}$. We did not find the statement in this form.

Ideas. The purity step reads: removing a subset of real codimension at least 3 does not change π_1. For $M_{g,\nu}$, the hypothesis of the last corollary (end and interior with the same Π_1) should follow from the simple connectivity of the curve complex (Harer).

Lean. Not attempted: no homotopy colimits of spaces or stratified spaces in Mathlib.

What would settle it. A reference for the gluing theorem, or a proof, and the check of the hypothesis for $M_{g,\nu}$.

7 Pseudoline arrangements

A pseudoline arrangement in the real projective plane is *bigonal along* one of its pseudolines D when D carries exactly two vertices of the arrangement, so that every other pseudoline meets D at one of them. In the page's notation, II_4 is four pseudolines in general position, II_5 five in general position, $\mathrm{II}'_{3,2}$ a pencil of three and a pencil of two without the line joining their centres, and III_6 the complete quadrangle (the six lines joining four points, no three collinear).

The page's boxed dichotomy (folder 80, pp. 20, 24, 28, from 1981), listed here as an open problem until 26 October 2026, has a short proof, and so does the numbering lemma of page 28; we record it so that the problem below keeps its context. Four pseudolines are in *general position* when no three of them pass through one point.

Claim. A bigon-free arrangement $\mathcal{A}$ either is II_4 or contains II_5, $\mathrm{II}'_{3,2}$ or III_6; and its pseudolines can be numbered, starting from a II_4 or a III_6, so that every partial arrangement is bigon-free.

Step 1: $\mathcal{A}$ contains four pseudolines in general position. $\mathcal{A}$ is not a pencil, so by Melchior's form of the Sylvester–Gallai theorem, valid for pseudolines [11, 10], it has a simple crossing $p = a \cap b$. Since a is not bigonal it carries vertices $q_1, \dots, q_k$ besides p, with $k \geq 2$. Take $c \neq a$ through q_i and $d \neq a$ through q_j, $i \neq j$. Then $\{a, b, c, d\}$ is in general position as soon as $c \cap b \neq d \cap b$: indeed c, d miss p, meet a at different points, and $c \cap d$ lies neither on a nor on b. If this fails for every choice, all these pseudolines meet b at one point r; two pseudolines meet only once, so each q_i lies on a and on a single other pseudoline c_i, all through r, and every

pseudoline meets a at p or at some q_i. Then $\mathcal{A} = \{a, b, c_1, \ldots, c_k\}$ and b carries only p and r, so it is bigonal, which is excluded.

Step 2: the dichotomy. Let $Q \subset \mathcal{A}$ be four pseudolines in general position, and $\ell \in \mathcal{A} \setminus Q$. If ℓ passes through none of the six vertices of Q, then $Q \cup \{\ell\}$ is a II_5; if through exactly one, a $\mathrm{II}'_{3,2}$. Otherwise ℓ passes through exactly two, which must be an opposite pair $\{x \cap y, z \cap w\}$ of the quadrilateral $Q = \{x, y, z, w\}$ (two vertices on a common side would force ℓ to be that side). Call such an ℓ a diagonal of Q; there are at most three. If $\mathcal{A}$ contains neither II_5 nor $\mathrm{II}'_{3,2}$, then $\mathcal{A}$ is Q plus some diagonals. One diagonal alone would be bigonal; two diagonals together with Q form a III_6 (the four triple points are the vertices of the quadrangle and the six pseudolines its sides); with three, $\mathcal{A}$ contains a III_6 again.

Step 3: the numbering. Adding pseudolines never removes a vertex from a pseudoline already there, so it suffices that each pseudoline, when added, meets the earlier ones in at least three points. A pseudoline meets the six sides of a III_6 in at least three points, since two vertices of III_6 never cover all six sides; so if $\mathcal{A} \supseteq \mathrm{III}_6$, start there and add the rest in any order. Otherwise $\mathcal{A}$ contains at most one diagonal δ of Q: start from Q, add every pseudoline other than δ (each meets Q in at least three points), and add δ last, when it has all its vertices in $\mathcal{A}$, at least three.

This agrees with a computer enumeration of all arrangements of at most nine pseudolines (`computations/bigon`). We have not found the statement in print; it is elementary, and probably folklore among people who work on arrangements.

Problem 14 (A bigon-free arrangement is determined by its skeleton)**.** Is a combinatorial arrangement of pseudolines in the real projective plane without bigonal pseudolines determined by its 1-skeleton together with the decomposition of the edges into pseudolines, functorially: is the functor from bigon-free arrangements to their decomposed skeleta fully faithful, so that every automorphism of the decomposed skeleton comes from the arrangement?

Source. folder 80, pp. 39–54. The statement of page 54 (« En résumé ») is clean. It composes a full faithfulness theorem (p. 39) for arrangements with an orientation of every non-concurrent triple (pp. 41–49, conditions (A1)–(A4)) with a canonical reconstruction of these orientations from the skeleton (pp. 49–54, (A5), (A6), with the complete quadrangle as residual case). Conditions (A1)–(A6) are said to hold « bien entendu » and not proved.

What is known. Local sequences determine an arrangement up to isomorphism in the Euclidean setting, and a Euclidean pseudoline arrangement graph has a unique planar embedding (Bose–Everett–Wismath); uniqueness of embeddings of graphs of large face-width is known (Robertson–Vitray, Mohar) [12]. We found no statement for the projective, bigon-free case with automorphisms.

Ideas. Our pointer, unchecked: the bigon exception coincides with the case where the map has face-width 2 (a non-contractible curve through the pencil centre and one point of the transversal), which is where uniqueness-of-embedding theorems stop applying. If bigon-free projective arrangements have face-width at least 3, Mohar–Thomassen's uniqueness theorems may give the result directly.

Lean. Not attempted: maps on surfaces are not in Mathlib.

What would settle it. A proof of (A1)–(A6), or the face-width bound with a uniqueness theorem.

8 How the list was made, and its limits

The fourteen problems were chosen from the register of findings among the rows still open and outside what Lean reached, and with a statement precise enough to be proved or refuted. Rows whose only interest is a slip on the page, a dating, or a statement now known to be in print were left out, as were the conjectures already in the published record (the standard conjectures, the section conjecture, the p-curvature conjecture, the homotopy hypothesis), which the companion paper discusses [17].

Three limits apply to every entry. The transcriptions are first passes by a language model, and doubtful readings are marked but not all checked against the facsimile; a statement may rest on a misread word, and the entries say where we know this. The literature searches were made by the same means, with full texts where they were open and abstracts otherwise; an entry that says « not found » reports a search, not an absence. And the steps marked « ours » have not been checked by a specialist.

Corrections, references and proofs are welcome at the address above. The register of findings, the Lean files and the notices of the folders are on the project's site, `https://grothendieck.commutator.io/`.

Declaration

This note was drafted with a language model (Claude Opus 5.5, Anthropic) from the edition's registers: the findings, the modernised readings and the Lean files. The transcriptions and the modernised readings it relies on were also made with language models (Claude Opus 5 and Opus 5.5). The author received no funding for this work.

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